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(54) **AUTOMATED STORAGE AND RETRIEVAL SYSTEM WITH FIRE DETECTION DEVICE AND METHODS OF LOCATING AND/OR VERIFYING FIRE OR SMOKE IN AN AUTOMATED STORAGE AND RETRIEVAL SYSTEM**

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See application file for complete search history.

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(56) **References Cited**

U.S. PATENT DOCUMENTS

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2017/0267452 A1* 9/2017 Goren B65G 1/0492
2019/0022441 A1* 1/2019 Konduri G05D 1/0094
(Continued)

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FOREIGN PATENT DOCUMENTS

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CN 202189434 U 4/2012
CN 107341947 A 11/2017
(Continued)

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OTHER PUBLICATIONS

(22) PCT Filed: **Apr. 6, 2020**

Office Action issued in counterpart Chinese Application No. 2020800309377 mailed on Feb. 22, 2024 (15 pages).

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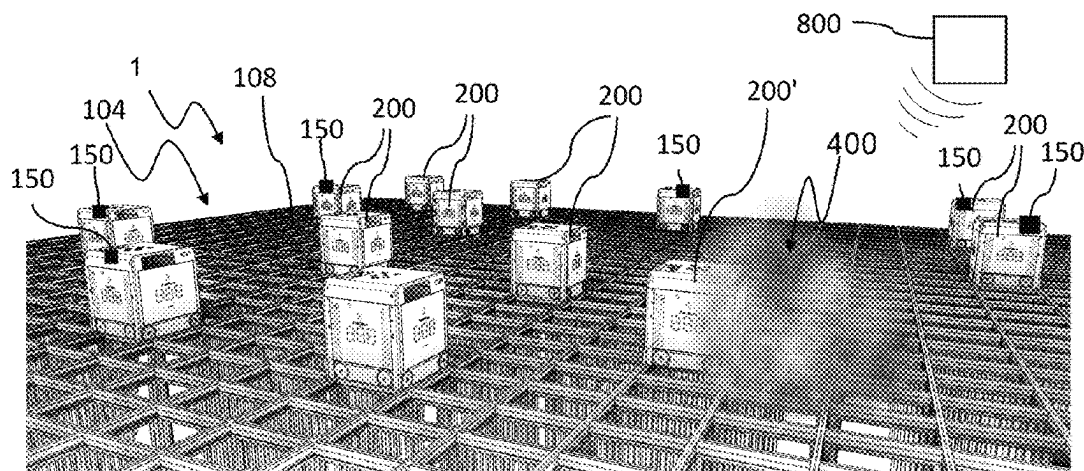
(52) **U.S. Cl.**

CPC **A62C 3/002** (2013.01); **B65G 1/0464** (2013.01)

(57) **ABSTRACT**

An automated storage and retrieval system includes a rail system, a master control system, and a plurality of remotely operated vehicles. The rail system includes a first and second set of parallel tracks arranged orthogonally in a horizontal plane. The master control system is configured to keep track of any remotely operated vehicle operating on the rail system. The plurality of remotely operated vehicles handling storage containers, which operate on the rail system, each include first and second sets of wheels for transport on the rail system, and a fire detection device configured to transmit data from the fire detection device to the master control system. The master control system includes a processing

(Continued)



device for processing the data from the fire detection devices so as to create a heat map of the automated storage and retrieval system.

15 Claims, 9 Drawing Sheets

(56)

References Cited

U.S. PATENT DOCUMENTS

2019/0233213	A1 *	8/2019	Phan-Quiroga	B65G 1/0492
2019/0240517	A1 *	8/2019	Clarke	B65G 1/0464
2019/0294165	A1 *	9/2019	Hofmann	A62C 3/0221
2019/0375588	A1 *	12/2019	Lindbo	B65G 1/0464

FOREIGN PATENT DOCUMENTS

CN	109230148	A	1/2019
JP	H09-221201	A	8/1997
JP	H09226909	A	9/1997
JP	2018-514483	A	6/2018
NO	317366	B1	10/2004
WO	2005010842	A1	2/2005
WO	2014075937	A1	5/2014
WO	2014090684	A1	6/2014
WO	2015193278	A1	12/2015
WO	2016/166353	A1	10/2016
WO	2016198467	A1	12/2016
WO	2017148963	A1	9/2017

OTHER PUBLICATIONS

Office Action issued in counterpart Japanese Application No. 2021-563016 mailed on Jan. 29, 2024 (5 pages).

J. Lenchner et al. "Towards Data Center Self-Diagnosis Using a Mobile Robot" Jan. 1, 2011; Retrieved from the Internet: URL :<https://www.researchgate.net/publication/221210942_Towards_data_center_self-diagnosis_using_a_mobile_robot> (11 pages).

International Search Report issued in International Application No. PCT/EP2020/059819, mailed May 13, 2020 (4 pages).

Written Opinion issued in International Application No. PCT/EP2020/059819; Dated May 13, 2020 (7 pages).

International Preliminary Report on Patentability issued in International Application No. PCT/EP2020/059819; mailed Jun. 25, 2021 (16 pages).

Office Action issued in Norwegian Application No. 20190546; Dated Nov. 20, 2019 (2 pages).

Hirokazu Moribayashi, Notice of Final Rejection for Japanese Patent Application No. P2021-563016, dated May 14, 2023, 5 pages, pub. by Japanese Patent Office.

Anonymous, First Office Action for Chinese Patent Application No. 202080030937.7, dated Jan. 14, 2025, 4 pages, pub. by SIPO, Beijing, China.

Dedek, Frederic, Extended European Search Report for European Patent Application No. 24196585.4, dated Feb. 17, 2025, 9 pages, pub. by the EPO, Munich, Germany.

Satya Veera Pavan Kumar Maddukuri et al, "A Low Cost Sensor Based Autonomous and Semi-Autonomous Fire-Fighting Squad Robot," Dec. 16, 2016, 2016 Sixth International Symposium on Embedded Computing and System Design (ISED), pp. 279-283, XP033117245, 5 pages.

M.L. NG, Requisition by examiner (Office action), pub. by Canadian Intellectual Property Office, Gatineau, Quebec, Mar. 27, 2025, 5 pages [all internally cited documents D1, D2, D3 have been previously cited and considered in this case].

* cited by examiner

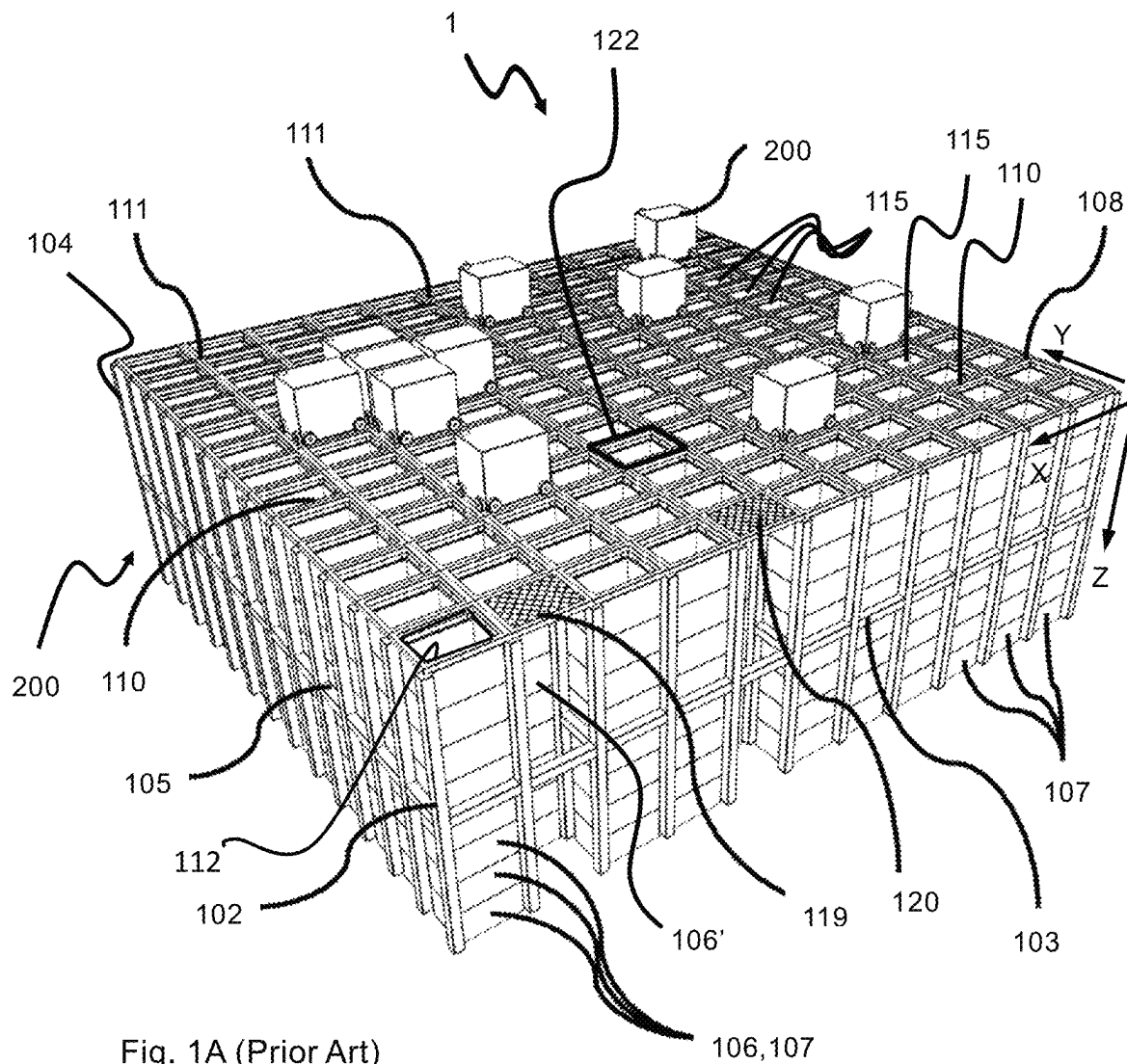


Fig. 1A (Prior Art)

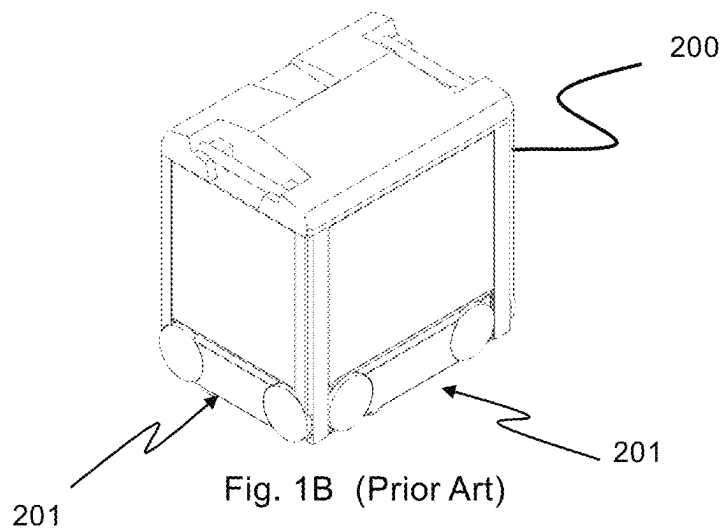
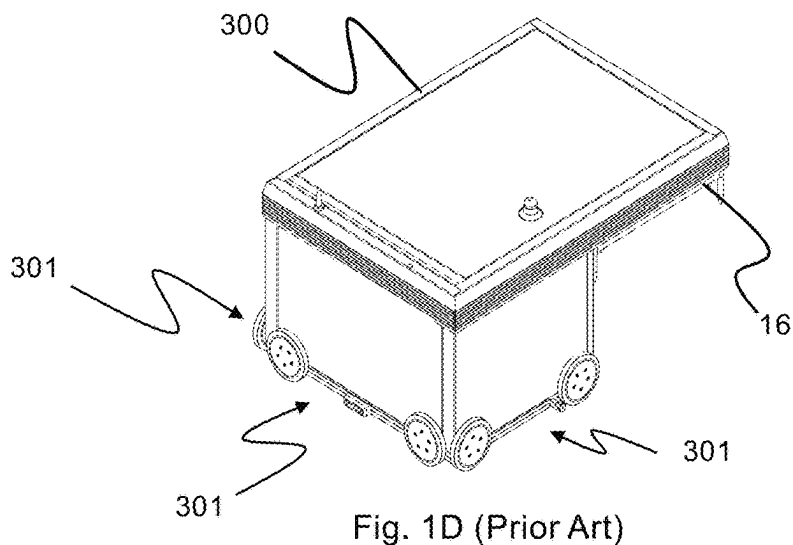
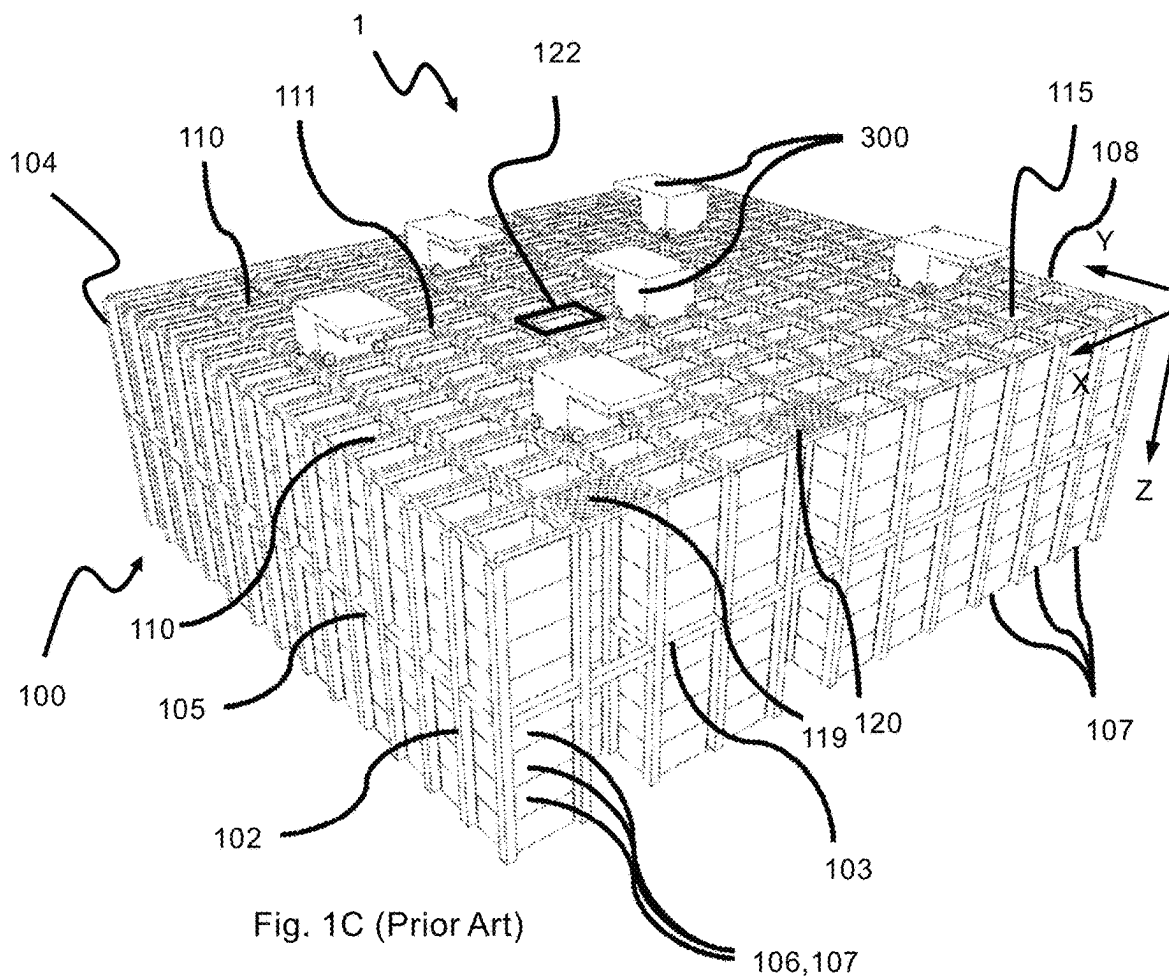


Fig. 1B (Prior Art)



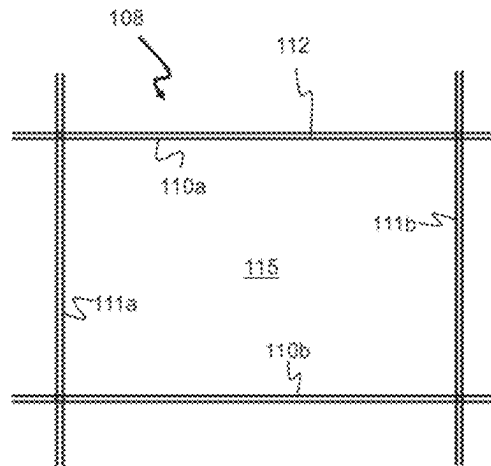


Fig. 2A (Prior Art)

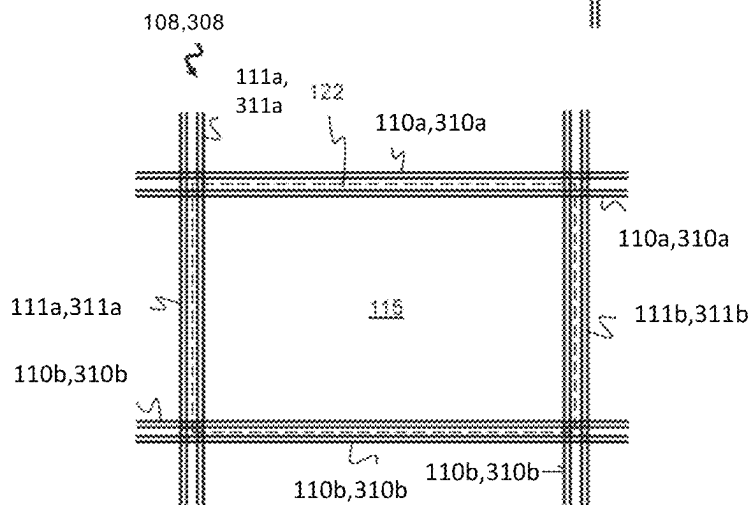


Fig. 2B (Prior Art)

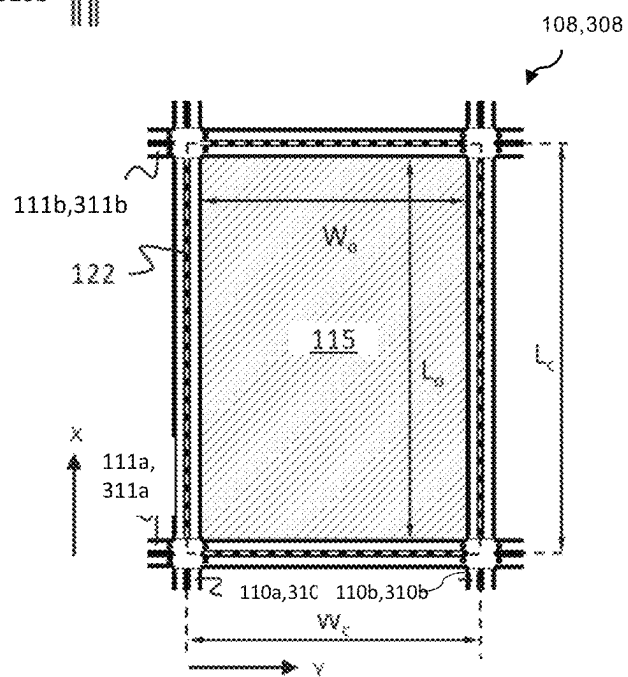


Fig. 2C (Prior Art)

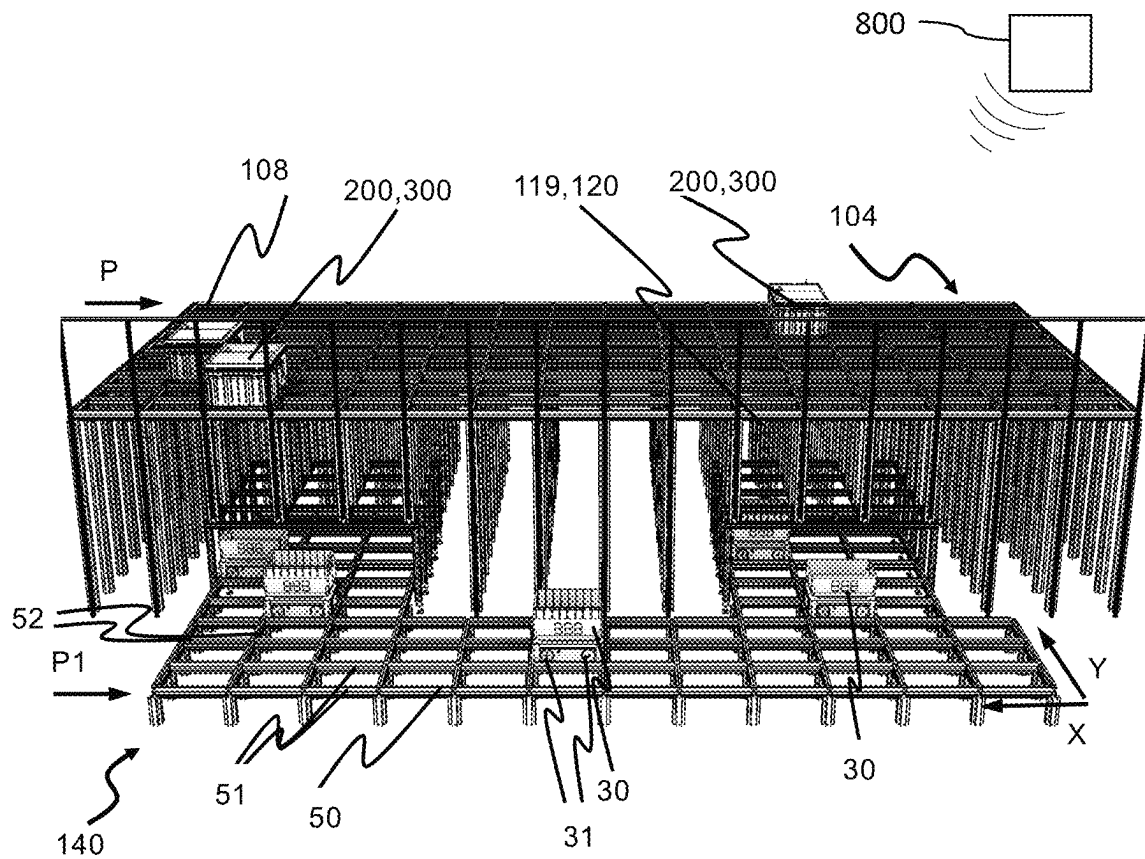


Fig. 3A

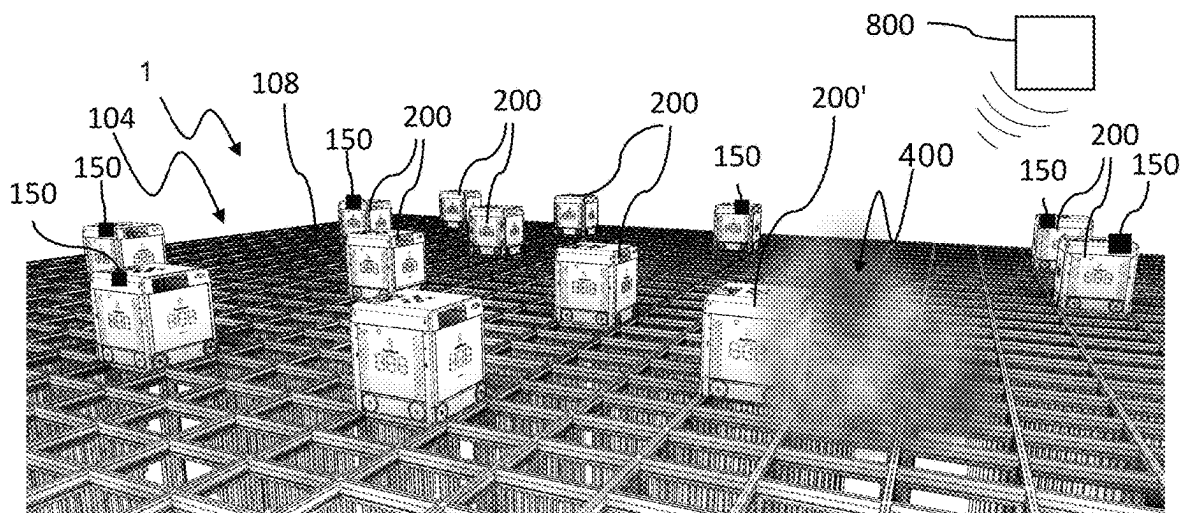


Fig. 3B

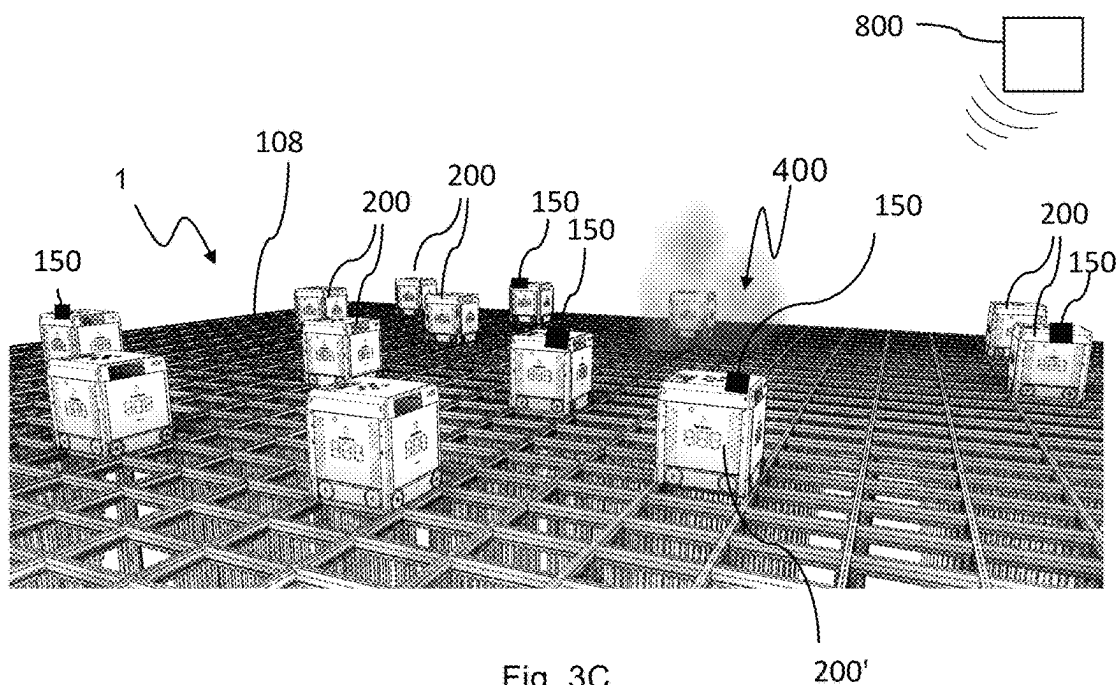


Fig. 3C

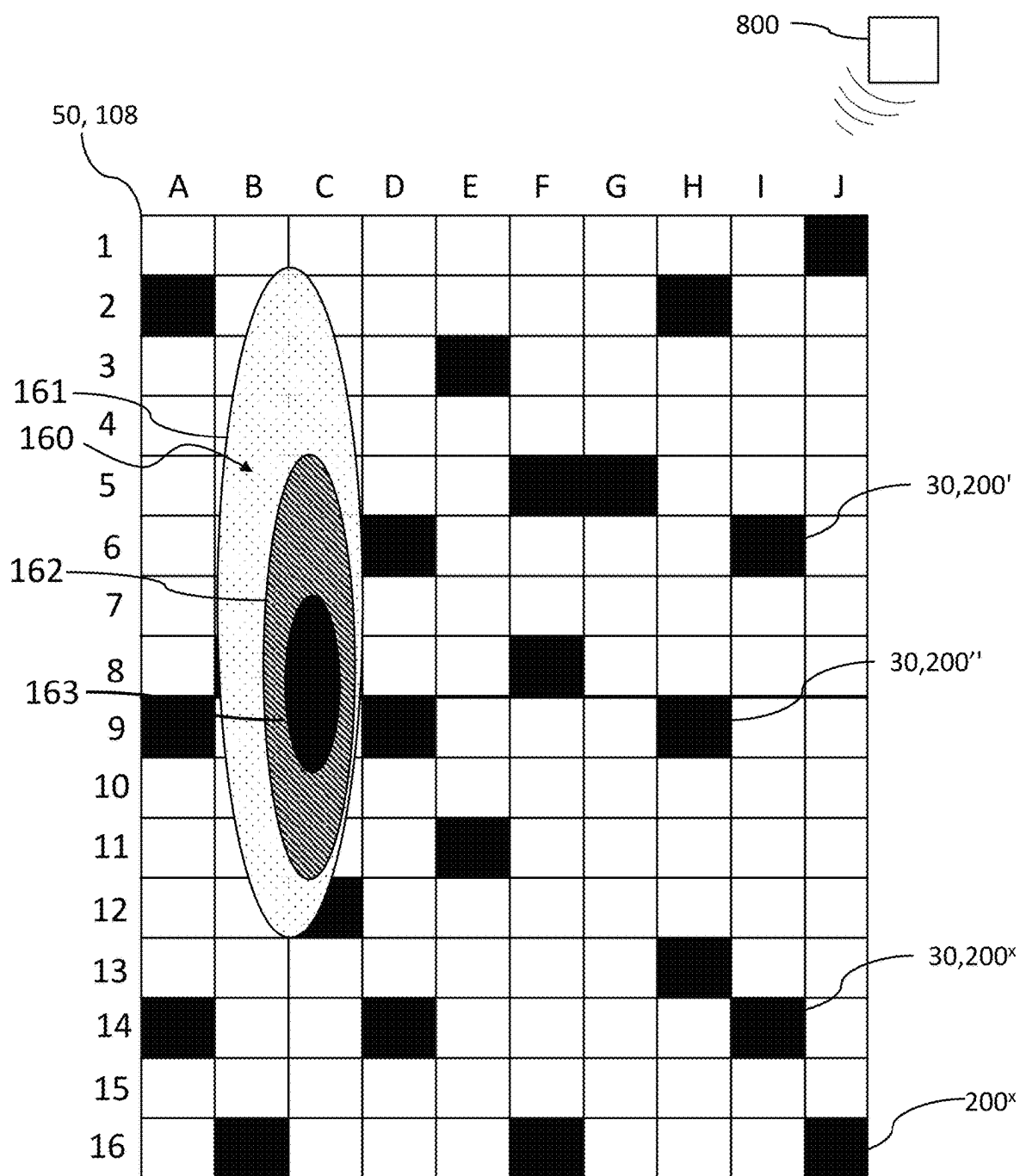


Fig. 4

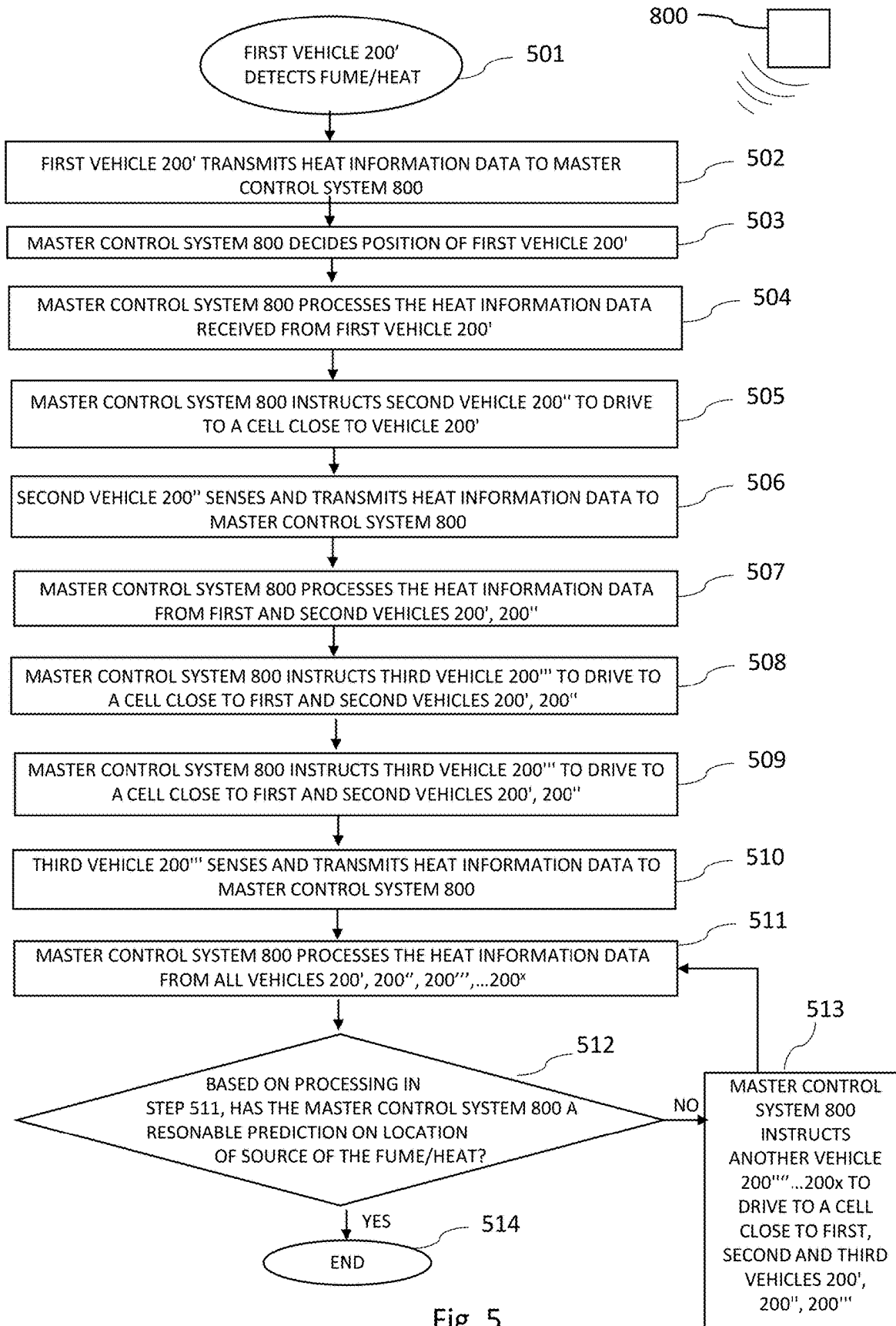


Fig. 5

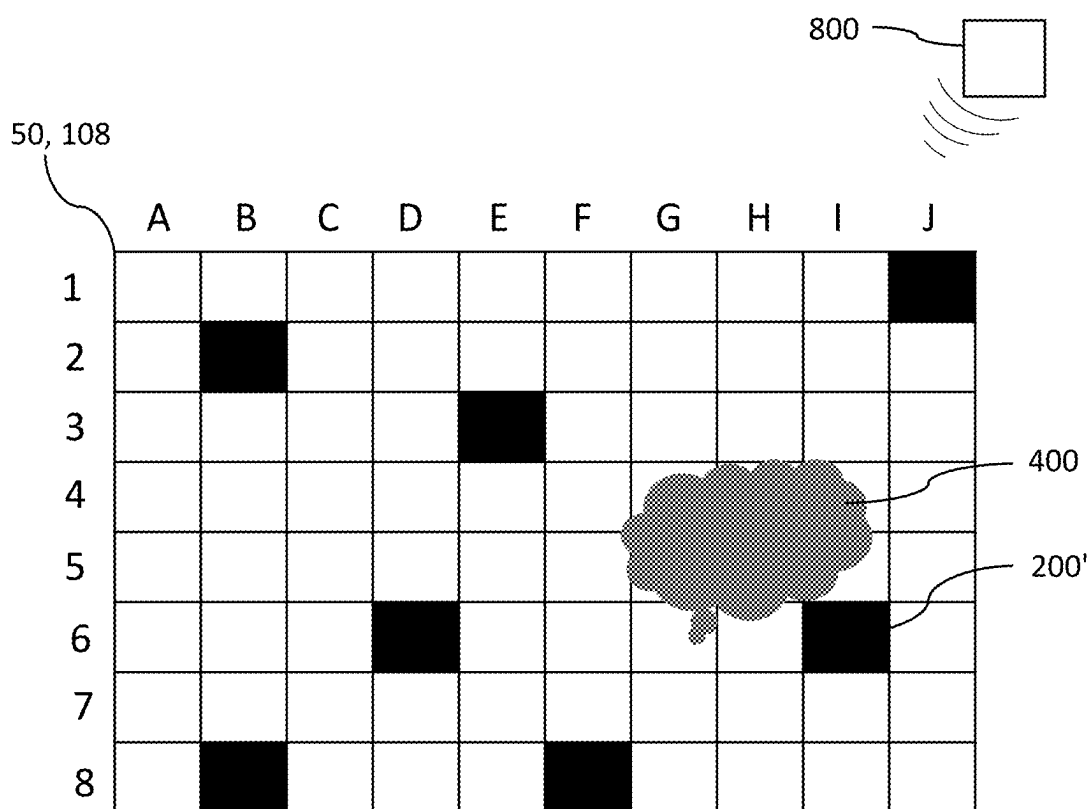


Fig. 6A

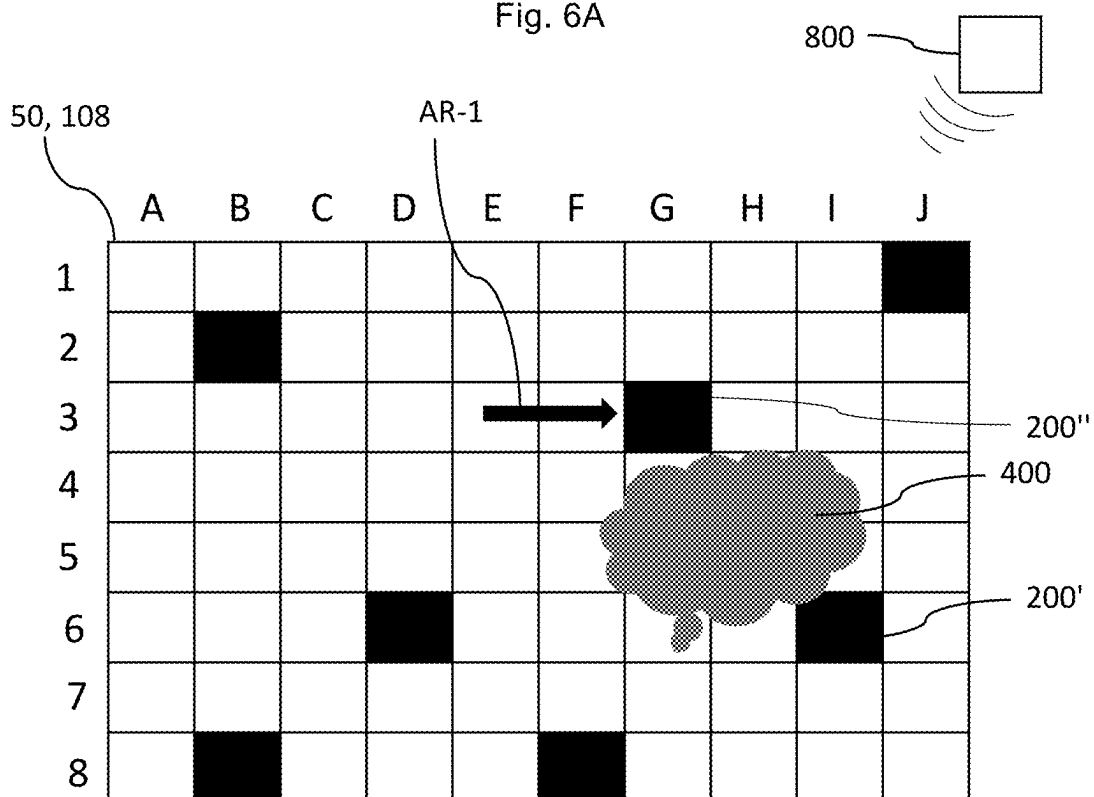


Fig. 6B

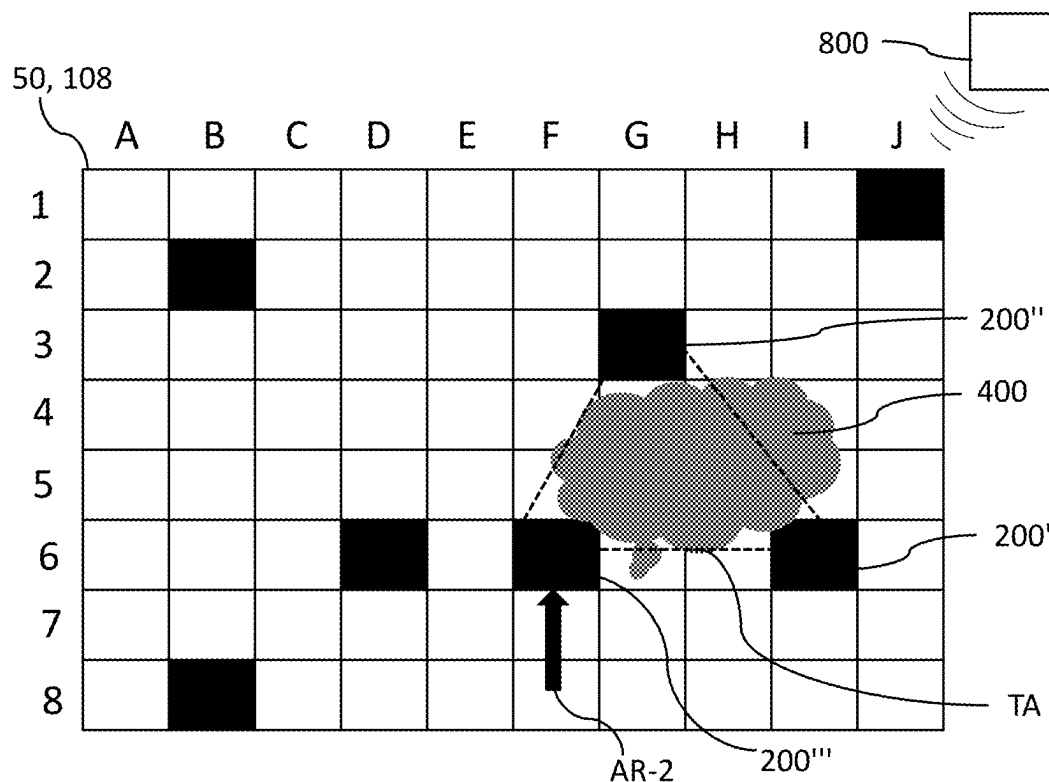


Fig. 6C

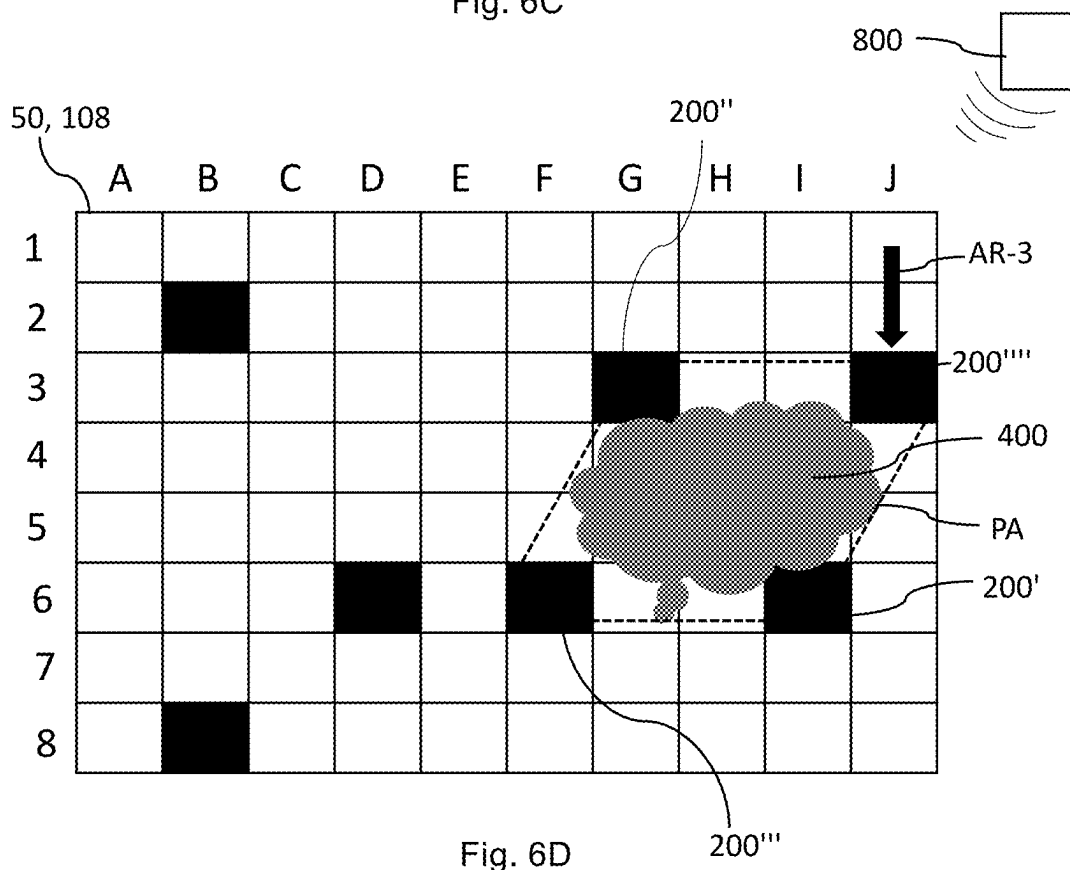


Fig. 6D

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AUTOMATED STORAGE AND RETRIEVAL SYSTEM WITH FIRE DETECTION DEVICE AND METHODS OF LOCATING AND/OR VERIFYING FIRE OR SMOKE IN AN AUTOMATED STORAGE AND RETRIEVAL SYSTEM

The present invention relates to an automated storage and retrieval system with fire detection device and methods of locating and/or verifying fire or smoke in an automated storage and retrieval system.

BACKGROUND AND PRIOR ART

FIGS. 1A and 1C disclose a typical prior art automated storage and retrieval system 1 with a framework structure 100. FIGS. 1B and 1D disclose prior art container handling vehicles 200, 300 operating on the system 1 disclosed in FIGS. 1A and 1C, respectively.

The framework structure 100 comprises a plurality of upright members 102 and optionally a plurality of horizontal members 103 supporting the upright members 102. The members 102, 103 may typically be made of metal, e.g. extruded aluminum profiles.

The framework structure 100 defines a storage grid 104 comprising storage columns 105 arranged in rows, in which storage columns 105 storage containers 106, also known as bins, are stacked one on top of another to form stacks 107.

Each storage container 106 may typically hold a plurality of product items (not shown), and the product items within a storage container 106 may be identical, or may be of different product types depending on the application.

The storage grid 104 guards against horizontal movement of the storage containers 106 in the stacks 107, and guides vertical movement of the storage containers 106, but does normally not otherwise support the storage containers 106 when stacked.

The automated storage and retrieval system 1 comprises a rail system 108 arranged in a grid pattern across the top of the storage grid 104, on which rail system 108 a plurality of container handling vehicles 200, 300 (as exemplified in FIGS. 1B and 1D) are operated to raise storage containers 106 from, and lower storage containers 106 into, the storage columns 105, and also to transport the storage containers 106 above the storage columns 105. The horizontal extent of one of the grid cells 122 constituting the grid pattern is in FIGS. 1A and 1C marked by thick lines.

Each grid cell 122 has a width which is typically within the interval of 30 to 150 cm, and a length which is typically within the interval of 50 to 200 cm. Each grid opening 115 has a width and a length which is typically 2 to 10 cm less than the width and the length of the grid cell 122 respectively due to the horizontal extent of the rails 110, 111.

The rail system 108 comprises a first set of parallel rails 110 arranged to guide movement of the container handling vehicles 200, 300 in a first direction X across the top of the frame structure 100, and a second set of parallel rails 111 arranged perpendicular to the first set of rails 110 to guide movement of the container handling vehicles 200, 300 in a second direction Y which is perpendicular to the first direction X. In this way, the rail system 108 defines grid columns above which the container handling vehicles 200, 300 can move laterally above the storage columns 105, i.e. in a plane which is parallel to the horizontal X-Y plane.

Each prior art container handling vehicle 200, 300 comprises a vehicle body and a wheel arrangement of eight wheels 201, 301 where a first set of four wheels enable the

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lateral movement of the container handling vehicles 200, 300 in the X direction and a second set of the remaining four wheels enable the lateral movement in the Y direction. One or both sets of wheels in the wheel arrangement can be lifted and lowered, so that the first set of wheels and/or the second set of wheels can be engaged with the respective set of rails 110, 111 at any one time.

Each prior art container handling vehicle 200, 300 also comprises a lifting device (not shown) for vertical transportation of storage containers 106, e.g. raising a storage container 106 from, and lowering a storage container 106 into, a storage column 105. The lifting device comprises one or more gripping/engaging devices (not shown) which are adapted to engage a storage container 106, and which gripping/engaging devices can be lowered from the vehicle 200, 300 so that the position of the gripping/engaging devices with respect to the vehicle 200, 300 can be adjusted in a third direction Z which is orthogonal the first direction X and the second direction Y.

Conventionally, and also for the purpose of this application, Z=1 identifies the uppermost layer of the storage grid 104, i.e. the layer immediately below the rail system 108, Z=2 the second layer below the rail system 108, Z=3 the third layer etc. In the exemplary prior art storage grid 104 disclosed in FIGS. 1A and 1C, Z=8 identifies the lowermost, bottom layer of the storage grid 104. Consequently, as an example, and using the Cartesian coordinate system X, Y, Z indicated in FIGS. 1A and 1D, the storage container identified as 106' in FIG. 1A can be said to occupy grid location or cell X=10, Y=2, Z=3. The container handling vehicles 101 can be said to travel in layer Z=0 and each grid column can be identified by its X and Y coordinates.

Each container handling vehicle 200 comprises a storage compartment or space (not shown) for receiving and stowing a storage container 106 when transporting the storage container 106 across the rail system 108. The storage space may comprise a container receiving space arranged centrally within the vehicle body, e.g. as is described in WO2014/090684A1, the contents of which are incorporated herein by reference.

Alternatively, the container handling vehicles 300 may have a cantilever construction, as is described in NO317366, the contents of which are also incorporated herein by reference.

The container handling vehicles 200 may have a footprint, i.e. an extent in the X and Y directions, which is generally equal to the lateral extent of a grid cell 122, i.e. the extent of a grid cell 122 in the X and Y directions, e.g. as is described in WO2015/193278A1, the contents of which are incorporated herein by reference. The term "lateral" used herein may mean "horizontal".

Alternatively, the container handling vehicles 200 may have a footprint which is larger than the lateral extent of (lateral area defined by) a grid column, e.g. as is disclosed in WO2014/090684A1.

The rail system 108 may be a single rail (also denoted single track) system, as is shown in FIG. 2A. Alternatively, the rail system 108 may be a double rail (also denoted double track) system, as is shown in FIG. 2B, thus allowing a container handling vehicle 200 having a footprint generally corresponding to the lateral area defined by a grid column 112 to travel along a row of grid columns even if another container handling vehicle 200 is positioned above a grid column neighboring that row. Both the single and double rail system, or a combination comprising a single and double rail arrangement in a single rail system 108, forms a grid pattern in the horizontal plane P comprising a plurality

of rectangular and uniform grid locations or grid cells **122**, where each grid cell **122** comprises a grid opening **115** being delimited by a pair of rails **110a**, **110b** of the first rails **110** and a pair of rails **111a**, **111b** of the second set of rails **111**. In FIG. 2B the grid cell **122** is indicated by a dashed box. For example, the sections of the rail-based system being made of aluminium are the rails, and on the upper surface of the rails, there are a pair of tracks that the wheels of the vehicle run in. However, the sections could be separate rails each with a track.

Consequently, rails **110a** and **110b** form pairs of rails defining parallel rows of grid cells running in the X direction, and rails **111a** and **111b** form pairs of rails defining parallel rows of grid cells running in the Y direction. Similarly, on a delivery rail system **50**, rails **51a** and **51b** form pairs of rails defining parallel rows of grid cells running in the X direction, and rails **52a** and **52b** form pairs of rails defining parallel rows of grid cells running in the Y direction.

As shown in FIG. 2C, each grid cell **122** has a width W_c , which is typically within the interval of 30 to 150 cm, and a length L_c which is typically within the interval of 50 to 200 cm. Each grid opening **115** has a width W_o and a length L_o which is typically 2 to 10 cm less than the width W_c and the length L_c of the grid cell **122**.

In the X and Y directions, neighboring grid cells are arranged in contact with each other such that there is no space therebetween.

In a storage grid **104**, a majority of the grid columns are storage columns **105**, i.e. grid columns **105** where storage containers **106** are stored in stacks **107**. However, a storage grid **104** normally has at least one grid column which is used not for storing storage containers **106**, but which comprises a location where the container handling vehicles **200**, **300** can drop off and/or pick up storage containers **106** so that they can be transported to a second location (not shown) where the storage containers **106** can be accessed from outside of the storage grid **104** or transferred out of or into the storage grid **104**. Within the art, such a location is normally referred to as a “port” and the grid column in which the port is located may be referred to as “port column” or “delivery column” **119**, **120**. The drop-off and pick-up ports of the rail system **108** where the container handling vehicles **200**, **300** operate are referred to as the “upper ports of a delivery column” **119**, **120**. While the opposite end of the delivery column is referred to as the “lower ports of a delivery column”.

The storage grids **104** in FIGS. 1A and 1C comprise two delivery columns **119** and **120**. The first delivery column **119** may for example comprise a dedicated drop-off port where the container handling vehicles **200**, **300** can drop off storage containers **106** to be transported through the delivery column **119** and further to an access or a transfer station, and the second delivery column **120** may comprise a dedicated pick-up port where the container handling vehicles **200**, **300** can pick up storage containers **106** that have been transported through the delivery column **120** from an access or a transfer station. Each of the ports of the first and second delivery column may comprise a port suitable for both pick up and drop of storage containers.

The second location may typically be a picking or a stocking station where product items are removed from or positioned into the storage containers **106**. In a picking or a stocking station, the storage containers **106** are normally never removed from the automated storage and retrieval system **1**, but are returned into the storage grid **104** once accessed. For transfer of storage containers out or into

the storage grid **104**, there are also lower ports provided in a delivery column, such lower ports are e.g. for transferring storage containers **106** to another storage facility (e.g. to another storage grid), directly to a transport vehicle (e.g. a train or a lorry), or to a production facility.

For monitoring and controlling the automated storage and retrieval system **1** (e.g. monitoring and controlling the location of respective storage containers **106** within the storage grid **104**; the content of each storage container **106**; and the movement of the container handling vehicles **200**, **300** so that a desired storage container **106** can be delivered to the desired location at the desired time without the container handling vehicles **200**, **300** colliding with each other), the automated storage and retrieval system **1** comprises a control system (not shown) which typically is computerized and which typically comprises a database for keeping track of the storage containers **106**.

A conveyor system comprising conveyors may be employed to transport the storage containers between the lower port of the delivery column and the access station.

If the lower port of the delivery column and the access station are located at different levels, the conveyor system may comprise a lift device for transporting the storage containers vertically between the port and the access station.

The conveyor system may be arranged to transfer storage containers between different grids, e.g. as is described in WO2014/075937A1, the contents of which are incorporated herein by reference.

Further, WO2016/198467A1, the contents of which are incorporated herein by reference, discloses an example of a prior art access system having conveyor belts (FIGS. 5a and 5b in WO2016/198467A1) and a frame mounted rail (FIGS. 6a and 6b in WO2016/198467A1) for transporting storage containers between delivery columns and work stations where operators can access the storage containers.

When a storage container **106** stored in the storage grid **104** disclosed in FIG. 1A is to be accessed, one of the container handling vehicles **200**, **300** is instructed to retrieve the target storage container **106** from its position in the storage grid **104** and to transport it to or through the delivery column **119**. This operation involves moving the container handling vehicle **200**, **300** to a grid location above the storage column **105** in which the target storage container **106** is positioned, retrieving the storage container **106** from the storage column **105** using the container handling vehicle's lifting device **16**, and transporting the storage container **106** to the delivery column **119**. If the target storage container **106** is located deep within a stack **107**, i.e. with one or a plurality of other storage containers positioned above the target storage container **106**, the operation also involves temporarily moving the above-positioned storage containers prior to lifting the target storage container **106** from the storage column **105**. This step, which is sometimes referred to as “digging” within the art, may be performed with the same container handling vehicle **200**, **300** that is subsequently used for transporting the target storage container **106** to the delivery column, or with one or a plurality of other cooperating container handling vehicles **200**, **300**. Alternatively, or in addition, the automated storage and retrieval system **1** may have container handling vehicles **200**, **300** specifically dedicated to the task of temporarily removing storage containers **106** from a storage column **105**. Once the target storage container **106** has been removed from the storage column **105**, the temporarily removed storage containers can be repositioned into the original

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storage column 105. However, the removed storage containers may alternatively be relocated to other storage columns 105.

When a storage container 106 is to be stored in the storage grid 104, one of the container handling vehicles 200, 300 is instructed to pick up the storage container 106 from the delivery column 120 and to transport it to a grid location above the storage column 105 where it is to be stored. After any storage containers positioned at or above the target position within the storage column stack 107 have been removed, the container handling vehicle 200, 300 positions the storage container 106 at the desired position. The removed storage containers may then be lowered back into the storage column 105, or relocated to other storage columns 105.

Over the years, few attempts have been made in order to improve the fire-safety in automated storage and retrieval systems as the ones described above. However, from WO 2017/148963 A1 it is known dedicated fire extinguishing service robot device on a grid. The disclosed service robot device is inactive during normal operation of the system and is only used in case of fire. The service robot device comprises fire extinguishing means and is capable of driving to any location on the grid in order to extinguish fire. Furthermore, the service robot device may comprise fire detecting means.

It is a problem with the prior art systems that they do not give the firefighting crew any information relating to where on the grid there is a heat- or fume emission.

One objective of the present invention is thus to provide a solution which provides the firefighting crew more reliable information with regards to location of any heat- or fume emission.

Another objective of the invention is to provide a solution which can verify whether a detection of heat- or fume emission is real.

SUMMARY OF THE INVENTION

The invention is set forth in the independent claims and the dependent claims describe alternatives of the invention.

It is described an automated storage and retrieval system comprising:

- a rail system comprising a first set of parallel tracks arranged in a horizontal plane and extending in a first direction, and a second set of parallel tracks arranged in the horizontal plane and extending in a second direction which is orthogonal to the first direction, which first and second sets of tracks form a grid pattern in the horizontal plane comprising a plurality of adjacent grid cells, each comprising a grid opening defined by a pair of neighboring tracks of the first set of tracks and a pair of neighboring tracks of the second set of tracks; and
- a master control system configured to keep track of any remotely operated vehicle operating on the rail system;
- a plurality of remotely operated vehicles operating on the rail system, each of the remotely operated vehicles comprising:
 - first and second sets of wheels for transport in the X and Y directions on the rail system and
 - a fire detection device configured to transmit data from the fire detection device to the master control system.

The master control system may comprise a processing device for processing the data from the fire detection devices such as to create a heat map of the automated storage and retrieval system. This may provide valuable input to the in order to assist the firefighting crew with regards to the

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probable source of fire or smoke, in particular as the automated storage and retrieval systems increase in size (up to 110x150 meters and above, and with 500000 bins and more). As the warehouses where the automated storage and retrieval systems are installed are often dark due to limited requirement of light sources in the warehouse during normal operation, and that any external power sources providing the light sources with electric power may shut-off in the event of fire, such temperature map or "heat map" can give valuable information in at least X and Y direction on where the source of heat or smoke/fume most likely has occurred.

The remotely operated vehicles may be container handling vehicles comprising a lifting assembly for picking up storage containers from the storage columns to a position above the lowest level of the transport mechanism, and the lifting assembly may comprise a lifting frame connectable to a storage container, the lifting frame being configured to lift and lower the storage containers from a position in the storage column to a position above the rail system.

The automated storage and retrieval system may further comprise:

- a plurality of stacks of storage containers arranged in storage columns located beneath the rail system, and each storage column may be located vertically below a grid opening.

A majority, i.e. more than 50% of the remotely operated vehicles may comprise an interface connection of the fire detection devices. The interface may be a plug-in module facilitating connection of the plug-in module to the respective remotely operated vehicles. The interface connection may be connected to a fire detection device.

It is further described a method of creating information with regards to location of any heat- or fume emission on an automated storage and retrieval system, the automated storage and retrieval system comprising:

- a rail system comprising a first set of parallel tracks arranged in a horizontal plane and extending in a first direction, and a second set of parallel tracks arranged in the horizontal plane and extending in a second direction which is orthogonal to the first direction, which first and second sets of tracks form a grid pattern in the horizontal plane comprising a plurality of adjacent grid cells, each comprising a grid opening defined by a pair of neighboring tracks of the first set of tracks and a pair of neighboring tracks of the second set of tracks; wherein the method comprises:

- operating a plurality of remotely operated vehicles on the rail system, each of the remotely operated vehicles being provided with a fire detection device, wherein the fire detection devices are configured to transmit data from the fire detection device to a master control system;

- operating a master control system, which master control system keeps track of any remotely operated vehicles operating on the rail system and is configured to receive input from any of the fire detection devices;

- utilizing the master control system to process any input from the fire detection device from any of the fire detection devices and using the master control system to generate a heat map which provide information of location of a source of the heat- or fume emission and/or the current temperature/temperature profile of the heat- or fume emission.

It is further described a method of verifying a detected heat- or fume indication on an automated storage and retrieval system, the automated storage and retrieval system comprising:

a rail system comprising a first set of parallel tracks arranged in a horizontal plane and extending in a first direction, and a second set of parallel tracks arranged in the horizontal plane and extending in a second direction which is orthogonal to the first direction, which first and second sets of tracks form a grid pattern in the horizontal plane comprising a plurality of adjacent grid cells, each comprising a grid opening defined by a pair of neighboring tracks of the first set of tracks and a pair of neighboring tracks of the second set of tracks; wherein the method comprises:

operating a plurality of remotely operated vehicles on the rail system, each of the remotely operated vehicles being provided with a fire detection device, wherein the fire detection devices are configured to transmit data from the fire detection device to a master control system;

operating a master control system, which master control system keeps track of any remotely operated vehicles operating on the rail system and is configured to receive data from the fire detection device from any of the fire detection devices; and in event of data from the fire detection device from a heat detection device on a first remotely operated vehicle indicating a detected heat- or fume emission, the method further comprises a step of: allocating a second remotely operated vehicle with a fire detection device to move to a cell nearby the position of the remotely operated vehicle which has identified heat- or fume to verify heat- or fume emission.

In case the second remotely operated vehicle provides data from the fire detection device to the master control system indicating presence of heat- or fume emission, the method may further comprise the steps of:

allocating a third remotely operated vehicle with a fire detection device to move to a cell nearby the position of the first and second remotely operated vehicles; and utilizing the master control system to process the data from the fire detection devices from the first, second and third remotely operated vehicles, and the master control system may, based on the processing of the data from the fire detection devices, decide whether a reasonable prediction on location of source of the fume- or heat emission can be given.

If the master control system has decided that a reasonable prediction on location of source of the fume- or heat emission can be given, the method may comprise the step of: utilizing the master control system to create a triangular arrangement enclosing the source of the fume, smoke or heat emission.

If the master control system has decided that a reasonable prediction on location of source of the fume- or heat emission cannot be given, the method may comprise the step of:

allocating a fourth remotely operated vehicle with a fire detection device to move to a cell nearby the position of the first, second and third remotely operated vehicles; and

utilizing the master control system to process the data from the fire detection devices from the first, second, third and fourth remotely operated vehicles, and the master control system may, based on the processing of the data from the fire detection devices, decide whether a reasonable prediction on location of source of the fume- or heat emission can be given.

If the master control system has decided that a reasonable prediction on location of source of the fume- or heat emission can be given, the method may comprise the step of:

utilizing the master control system to create a four-cornered polygon enclosing the source of the fume, smoke or heat emission.

The fire detection devices may be fume or smoke detectors and the data from the fire detection devices may represent presence of fume or smoke.

The fire detection devices may be heat detectors and the data from the fire detection devices may represent presence of heat.

The fire detection devices may be a combination of fume or smoke detectors and heat detectors and the data from the fire detection devices may represent presence of fume or smoke or heat. In this latter aspect, increased probability of locating the actual source of the emission of fume, smoke or heat is provided, as the master control system may combine different kinds of data from the heat detection devices (e.g. fume, smoke and heat) to produce an even more reliable heat map or predicted location of the source.

The fire detection device may be smoke detectors of ionization (ionic) type or photoelectric (optic) type or a combination of ionization or photoelectric type. Furthermore, as indicated above, the fire detection devices may be combinations of multiple sensor types such as heat detection, optical smoke detection, carbon monoxide detection creating rapid acting and fail-proof multi-sensor smoke detectors to reduce false alarms whilst giving the earliest warning of fire. The fire detection devices may further or additionally comprise infra-red sensors, cameras, CO concentration meters, CO₂ concentration meters, temperature sensors, opacity meters etc.

Opacity meters detect and measure the amount of light blocked and is defined as the measure of the impenetrability of electromagnetic radiation through a sample. The opacity instruments are normally used to analyse air and smoke which media types can be described as “clean” or “dirty” dependent on their opacity. Opacity measurement largely depends on the concentration of particles within a sample. When electromagnetic radiation—in this case visible light or infrared radiation—is emitted toward a sample, suspended solids will scatter, reflect, and absorb the radiation, preventing it from passing through. Opacity sensors measure the quantity of transmitted light and convert it to a relevant value.

The method may further comprise, prior to the last step, a step of:

using a stationary fire detection device arranged in or at the automated storage and retrieval system. The stationary fire detection devices may be arranged on or within the storage grid or delivery system, or adjacent the storage grid or delivery system.

The remotely operated vehicles are normally widely spread on the grid, thus if at least a majority of the remotely operated vehicles are provided with a fire detection device, there is a high possibility of early detection of heat- or fume emission. Therefore, it is not necessarily necessary that all of the remotely operated vehicles are provided with a fire detection device. Furthermore, as the master control system keeps track of the position of all of the remotely operated vehicles (i.e. both any container handling vehicles and delivery vehicles), if any heat- or smoke detection device senses smoke or heat, the master control system automatically knows the location of the remotely operated vehicle (and thus the location of the heat- or fume emissions).

An advantage of providing the remotely operated vehicles with heat- or smoke detection device(s) internally or externally of the remotely operated vehicle is earlier identifica-

tion of potential heat- or fume emission compared to solutions with heat- or smoke detectors in the ceiling.

In an aspect, if the master control system receives data representing heat- or smoke information it may utilize the system's so-called XHandler ("exception-handler") program. The XHandler program autonomously handles specific error exceptions in an automatic storage and retrieval system without shutting down operations. For example, if a container handling vehicle loses its position, XHandler will block a safe area around the container handling vehicle and help determine the location by identifying the nearby bin pattern. If XHandler can't determine this, another container handling vehicle will make a unique bin pattern (depth of several bins). If it can't, another container handling vehicle will make a unique bin pattern by removing a few bins.

In the present case, if the master control system receives data representing heat- or smoke or fume information, it may employ the XHandler program to instruct other container handling vehicles or delivery vehicles to the location where heat or smoke has been detected in order to confirm whether there is actually a heat- or fume emission which needs to be taken care of or if there is a false alarm.

False alarms may occur in many different events, such as due to short sparks or ignition resulting from friction between the wheels of the remotely operated vehicle and the tracks, internal heating inside a remotely operated vehicle due to overload, motor failure, lifting device motor failure etc. in the remotely operated vehicle.

The heat- or smoke detection devices may at least transmit data from the fire detection device indicating a heat- or fume emission to a master control system.

The fire detection devices may comprise combined transmitters and receivers, such that they are able to transmit and receive data from the fire detection device to the master control system. The transmitters and receivers of the fire detection devices can be connected to a controller, which controller may be similar to the ones which are used for general communication between the master control system and the remotely operated vehicles. Alternatively, the controllers for the transmitters and receivers may be different from the ones which are used for the general communication. The skilled person will know how to program such controllers to communicate with the master control system, thus this will not be further elaborated on herein. In addition, it may also be possible that the fire detection devices are able to transmit data from the fire detection device to other fire detection device(s) and receive data from the fire detection device from other fire detection device(s).

The main grid may be a storage grid and the plurality of remotely operated vehicles may be container handling vehicles operating on the storage grid, which container handling vehicles comprise a lifting device for lifting and lowering storage containers from below to a position above the uppermost point of the rail system of the storage grid.

Alternatively, the main grid may be a delivery grid and the plurality of remotely operated vehicles may be delivery vehicles. The remotely operated vehicles may be delivery vehicles comprising a closed bottom portion and may be configured to receive storage containers from above. In contrast to the container handling vehicles operating on the storage grid, which container handling vehicles comprise lifting device for lifting and lowering storage containers from below, the delivery vehicles comprises a closed bottom portion and are configured to receive storage containers from above (for example from container handling vehicles operating on storage grid at a level above the delivery system where the delivery vehicles operate).

If a plurality of the remotely operated vehicles are provided with a fire detection device, input in the form of data from the fire detection device from two or more fire detection devices on at least two remotely operated vehicles, can be processed in order to generate a "heat map" which can provide the firefighting crew with valuable information in terms of location of the fire or smoke and/or the current temperature/temperature profile of the heat- or fume emission. In order to obtain a more reliable heat map than heat maps based on fire detection devices providing only "yes/no" input with regards to presence of heat- or fume, it may be advantageous to utilize three or more fire detection devices which can provide additional input to the master control system with regards to concentration of fume or smoke, or in case of heat detection, additional input from e.g. an infra-red camera or temperature sensor on the remotely operated vehicle.

Some or all of the remotely operated vehicles may comprise an interface connection of the fire detection devices. The interface may be a plug-in module facilitating connection of the plug-in module to the respective remotely operated vehicles.

The remotely operated vehicles may be equipped with light in order to assist the firefighting crew, and in particular in case of loss of power supply shutting down normal light sources.

BRIEF DESCRIPTION OF THE DRAWINGS

The following drawings depict exemplary embodiments of the present invention and are appended to facilitate the understanding of the invention. However, the features disclosed in the drawings are for illustrative purposes only and shall not be interpreted in a limiting sense.

FIG. 1 A-D are perspective views of a prior art automated storage and retrieval system, where FIG. 1A and FIG. 1C show the complete system and FIG. 1B and FIG. 1D show examples of a system operable with prior art container handling vehicles;

FIG. 2A-C is a top view of a container handling vehicle rail system, where FIG. 2A shows a single rail system, FIG. 2B shows a double rail system and FIG. 2C shows a double rail system with the width and length of a container handling vehicle grid cell indicated;

FIG. 3A is a perspective view of an exemplary embodiment of a storage grid and a delivery system comprising delivery rail system and delivery vehicles;

FIGS. 3B and 3C show two examples where fume, smoke or heat emission has occurred in an automated storage and retrieval system;

FIG. 4 shows an example of a heat map created by the master control system based on input from a plurality of fire detection devices arranged on container handling vehicles;

FIG. 5 is a flow-chart of an example of steps to be taken in the event a container handling vehicle detects fume, smoke or heat;

FIGS. 6A-6D are sequential step-by-step illustrations of the flow-chart of FIG. 5, where one and one container handling vehicle with a fire detection device is instructed to drive to grid cells close to a detected fume, smoke or heat in order for the master control system to process data from the fire detection device received from the container handling vehicles to predict a location of the source of the fume, smoke or heat.

DETAILED DESCRIPTION OF THE INVENTION

In the following, different alternatives will be discussed in more detail with reference to the appended drawings. It

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should be understood, however, that the drawings are not intended to limit the scope of the invention to the subject-matter depicted in the drawings. Furthermore, even if some of the features are described in relation to the system only, it is apparent that they are valid for the related methods as well, and vice versa.

With reference to FIGS. 1A-D the storage grid **104** of each storage structure **1** constitutes a framework **100** of in total 143 grid columns **112** (see grid column **112** in top front corner, i.e. grid location or cell $X=11, Y=1, Z=0$), where the width and length of the framework corresponds to the width and length of 13 and 11 grid columns **112**, respectively. The top layer of the framework **100** is a container handling vehicle rail system/rail system **108** on which a plurality of container handling vehicles **200, 300** are operated.

The framework **100** of the storage system **1** is constructed in accordance with the above mentioned prior art framework **100** described above, i.e. a plurality of upright members **102** and a plurality of horizontal members **103** which are supported by the upright members **102**, and further that the horizontal members **103** includes a container handling vehicle rail system **108** of parallel rails **110, 111** in the X direction and the Y direction, respectively, arranged across the top of storage columns **105**. The horizontal area of a single grid opening **115**, i.e. along the X and Y directions, may be defined by the distance between adjacent rails **110** and **111**, respectively (see also FIGS. 2A-2C). In FIGS. 1A and 1C, a grid cell **122** is marked on the rail system **108** by thick lines. The area between adjacent rails is the grid opening **115**—the grid cell **122** is the area to the outer side edges of the opposing rails running along the opposite sides of the grid opening. Alternatively, if these are defined in terms of double-tracked rails, then the area is to the midpoint or centre of each adjacent rail.

The container handling vehicle rail system **108** allows the container handling vehicles **200, 300** to move horizontally between different grid locations, where each grid location is associated with a grid cell **122**.

In FIGS. 1A and 1C the storage grid **104** is shown with a height of eight cells. It is understood, however, that the storage grid **104** can in principle be of any size. In particular it is understood that storage grid **104** can be considerably wider and/or longer than disclosed in FIGS. 1A and 1C. For example, the storage grid **104** may have a horizontal extent of more than 700×700 grid cells **122**, or any size in between these examples, e.g. 100×100 grid cells, 200×200 grid cells, 500×500 grid cells etc. Also, the grid **104** can be considerably deeper than disclosed in FIGS. 1A and 1C. For example, the storage grid **104** may be more than twelve grid cells deep.

The storage grid **104** is equal or similar to the prior art storage grid **104** as described above, i.e. a storage grid **104** comprising a rail system **108**; a plurality of stacks **107** of storage containers **106**, a plurality of container handling vehicles **300** for lifting and moving storage containers **106** stacked in the stacks **107** and a delivery column **119, 120** configured to receive a storage container **106** from a container handling vehicle **200, 300**.

The rail system **108** comprises a first set of parallel rails **110** arranged in a horizontal plane (P) and extending in a first direction (X) and a second set of parallel rails **111** arranged in the horizontal plane (P) and extending in a second direction (Y) which is orthogonal to the first direction (X). The first and second sets of rails **110, 111** form a grid pattern in the horizontal plane (P) comprising a plurality of adjacent grid cells **122**.

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Each grid cell **122** displays a grid opening defined by a pair of neighbouring rails of the first set of rails **110** and a pair of neighbouring rails of the second set of rails **111**.

The plurality of stacks **107** are arranged in storage columns **105** located beneath the rail system **108**, wherein each storage column **105** is located vertically below a grid cell **122**.

Each container handling vehicle **200, 300** is configured to move on the rail system **108** above the storage columns **105**.

The storage container vehicles **200, 300** may be of any type known in the art, e.g. any one of the automated container handling vehicles disclosed in WO2014/090684 A1, in NO317366 or in WO2015/193278A1.

The rail system **108** may be a single rail system, as is shown in FIG. 2A. Alternatively, the rail system **108** may be a double rail system, as is shown in FIG. 2B. In a further alternative, the rail system **108** may be a combination of single and double rail system. Details of the single and double rail systems are disclosed in this specification under the section of background and prior art.

Perspective views of an automated storage and retrieval system are shown in FIG. 3A. The automated storage and retrieval system comprises a storage grid **104**, onto which a plurality of container handling vehicles **200, 300** operates, and a delivery system **140** comprising a delivery rail system **50** onto which a plurality of delivery vehicles **30** operates. The delivery vehicles **30** are provided with wheel arrangement **31** comprising first and second sets of wheels for driving the delivery vehicles in the first direction (X) and the second direction (Y). A master control system **800** keeps track of the container handling vehicles **200, 300** operating on the storage grid **104** and the delivery vehicles **30** operating on the delivery rail system **30**.

Further, the delivery system **140** comprises one or more of the delivery vehicles **30** as described above, i.e. delivery vehicles **30** configured to receive and support one or more storage containers **106** for transport between one or more delivery columns **119, 120** of the storage grid **104** and one or more predetermined positions outside the storage grid **104**. The predetermined positions may for example be a second location, a container accessing station, a conveyor line, another storage container, or a transport vehicle such as a truck.

The delivery system **140** may further comprise a delivery rail system **50** situated below a delivery port of the one or more delivery columns **119, 120**.

As shown in FIG. 3, the delivery rail system **50** may be constructed in the same way or a similar way as the rail system **108** for the container handling vehicles **200, 300**.

Hence, the delivery rail system **50** may comprise a first set of parallel rails **51** arranged in a horizontal plane (P1) and extending in a first direction (X), and a second set of parallel rails **52** arranged in the horizontal plane (P1) and extending in a second direction (Y) which is orthogonal to the first direction (X).

The delivery rail system **50** may also be a double rail system, as is shown in FIG. 2B, thus allowing a delivery vehicle **30** having a footprint generally corresponding to the lateral area defined by a delivery grid column to travel along a row of grid columns even if another delivery vehicle **30** is positioned above a grid column neighbouring that row.

Both the single and double rail system, or a combination comprising a single and double rail arrangement in a single rail system, forms a grid pattern in the horizontal plane P1 comprising a plurality of rectangular and uniform grid locations or grid cells, where each grid cell comprises a grid

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opening being delimited by a pair of rails of the first rails and a pair of rails of the second set of rails.

In contrast to the container handling vehicles **200**, **300** operating on the storage grid **104**, which container handling vehicles **200**, **300** comprise a lifting device **16** for lifting and lowering storage containers **106** from below, the delivery vehicles **30** comprises a closed bottom portion and are configured to receive storage containers **106** from above (for example from container handling vehicles **200**, **300** operating on storage grid **104** at a level above the delivery system **140** where the delivery vehicles **30** operate).

FIGS. 3B and 3C show two examples where fume, smoke or heat emission **400** has occurred on a storage grid **104** in an automated storage and retrieval system **1**. A plurality of container handling vehicles **200**, **200'** operates on a rail system **108** of a storage grid **104** in an automated storage and retrieval system **1** and communicates with a master control system **800**. Some of the container handling vehicles **200**, **200'** have been equipped with fire detection device **150**. However, preferably a majority, preferably above 50%, of the container handling vehicles **200**, **200'** comprises a fire detection device **150**, possibly together with stationary fire detection devices **150** arranged on the storage grid **104**.

It is clear that, although not disclosed, that the delivery vehicles **30** operating on a delivery rail system **50** of a delivery system **140** like the one in FIG. 3A, may also be equipped with, or comprise, fire detection device(s) **150** in order to detect any fume, smoke or heat emission **400** within the delivery system **140**, possibly together with stationary fire detection devices **150** arranged on the delivery system **140**.

FIG. 4 shows an example of a heat map created by the master control system based on input from a plurality of fire detection devices **150** arranged on container handling vehicles **200**, **300**.

In the following, an example of utilizing fire detection devices **150** on container handling vehicles **200'**, **200^x** such as to create a heat map **160** will be described in greater detail. However, it is apparent, as is also shown in FIG. 4, fire detection devices **150** may also be arranged on delivery vehicles **30** such that a similar heat map may be created based on data from the fire detection device from fire detection devices **150** on delivery vehicles **30**. Order for the master control system **800** to provide the heat map **160**, preferably three or more fire detection devices **150** which can provide additional input to the master control system **800** with regards to concentration of fume or smoke, or in case of heat detection, additional input from e.g. an infra-red camera or temperature sensor on the container handling vehicle **200'**, **200^x**, are utilized. Then, based on the location of the container handling vehicle **200'**, **200^x** on the rail system **108** and the concentration of fume or smoke measured by the heat detection device **150**, the master control system **800** can create a heat map **160** which indicates the probable location of the source of the fume, smoke or heat emission (see FIGS. 3B and 3C). The disclosed heat map **160** comprises an outer portion **161** indicated by coarse-grains, an intermediate portion **161** indicated by stripes and an inner portion **162** indicated by solid black. The heat map **160** may indicate that it is certain that the source of the fume, smoke or heat emission **400** is within the outer portion **161**, and almost certain that the source of the fume, smoke or heat emission **400** is within the striped portion **161**, and very likely that the source of the fume, smoke or heat emission **400** is within the inner portion **162**. If the master control system **800** has drawn up such a heat map **160**, it is most promising for any fire-fighting crew to focus their initial fire-fighting in the

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inner portion **162** (i.e. within cell locations B7-B10 and C7-C10) on FIG. 4. Although the figure is shown in relation to container handling vehicles on a storage grid, a similar figure may be made based on input from fire detection devices arranged on delivery vehicles operating on a delivery rail.

FIG. 5 is a flow-chart of an example of steps to be taken in the event a container handling vehicle or delivery vehicle detects fume, smoke or heat. Such a process may include the following steps:

Step **501**: VEHICLE DETECTS FUME/HEAT
 Step **502**: FIRST VEHICLE **200'** TRANSMITS DATA FROM THE FIRE DETECTION DEVICE TO MASTER CONTROL SYSTEM **800**
 Step **503**: MASTER CONTROL SYSTEM **800** DECIDES POSITION OF FIRST VEHICLE **200'**
 Step **504**: MASTER CONTROL SYSTEM **800** PROCESSES THE DATA FROM THE FIRE DETECTION DEVICE RECEIVED FROM FIRST VEHICLE **200'**
 Step **505**: MASTER CONTROL SYSTEM **800** INSTRUCTS SECOND VEHICLE **200''** TO DRIVE TO A CELL CLOSE TO VEHICLE **200'**
 Step **506**: SECOND VEHICLE **200''** SENSES AND TRANSMITS DATA FROM THE FIRE DETECTION DEVICE TO MASTER CONTROL SYSTEM **800**
 Step **507**: MASTER CONTROL SYSTEM **800** PROCESSES THE DATA FROM THE FIRE DETECTION DEVICE FROM FIRST AND SECOND VEHICLES **200'**, **200''**
 Step **508**: MASTER CONTROL SYSTEM **800** INSTRUCTS THIRD VEHICLE **200'''** TO DRIVE TO A CELL CLOSE TO FIRST AND SECOND VEHICLES **200'**, **200''**
 Step **509**: MASTER CONTROL SYSTEM **800** INSTRUCTS THIRD VEHICLE **200'''** TO DRIVE TO A CELL CLOSE TO FIRST AND SECOND VEHICLES **200'**, **200''**
 Step **510**: THIRD VEHICLE **200'''** SENSES AND TRANSMITS DATA FROM THE FIRE DETECTION DEVICE TO MASTER CONTROL SYSTEM **800**
 Step **511**: MASTER CONTROL SYSTEM **800** PROCESSES THE DATA FROM THE FIRE DETECTION DEVICE FROM ALL VEHICLES **200'**, **200''**, **200'''**, . . . **200^x**
 Step **512**: BASED ON PROCESSING IN STEP **511**, HAS THE MASTER CONTROL SYSTEM **800** A REASONABLE PREDICTION ON LOCATION OF SOURCE OF THE FUME/HEAT?
 IF "YES" IN STEP **512** proceed to step **514**: END
 IF "NO" IN STEP **512**: proceed to step **513**: MASTER CONTROL SYSTEM **800** INSTRUCTS ANOTHER VEHICLE **200''''** . . . **200^x** TO DRIVE TO A CELL CLOSE TO FIRST, SECOND AND THIRD VEHICLES **200'**, **200''**, **200'''**

FIGS. 6A-6D are sequential step-by-step illustrations of the flow-chart of FIG. 5, where one and one container handling vehicle **200' . . . ^x** with a fire detection device is instructed to drive to grid cells close to a container handling vehicle **200'** that has detected fume, smoke or heat **400** in order for the master control system **800** to process data from the fire detection device received from the container handling vehicles **200' . . . ^x** to predict location of the source of the fume, smoke or heat **400**.

In FIG. 6A a first container handling vehicle **200'** in cell H5, which first container handling vehicle **200'** is provided with a fire detection device (not shown in FIG. 6A), detects fume, smoke or heat **400**. The first container handling

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vehicle **200'** transmits data from the fire detection device to master control system **800**. The master control system **800** keeps continuous track of all of the container handling vehicles **200'**, and thus know the position of the first container handling vehicle **200'** which has detected fume, smoke or heat **400**. The master control system **800** processes the data from the fire detection device received data from the fire detection device received from the first container handling vehicle **200'**.

In FIG. 6B the master control system **800**, based on the input from the first container handling vehicle **200'**, has instructed a second container handling vehicle **200''** initially positioned in cell E3, to move to cell G3 (as illustrated by arrow AR-1) which is closer to the first container handling vehicle **200'** (and expected to be closer to the source of the fume, smoke or heat emission **400**). When the second container handling vehicle **200''** enters or has entered cell G3, the fire detection device of the second container handling vehicle **200''** transmits data from the fire detection device to the master control system **800**. The master control system **800** processes data from the fire detection device from the first and second container handling vehicles **200'**, **200''**.

In FIG. 6C the master control system **800**, based on the input from the first and second container handling vehicles **200'**, **200''**, has instructed a third container handling vehicle **200'''** initially positioned in cell F8, to move to cell F6 (as illustrated by arrow AR-2) which is closer to the first and second container handling vehicles **200'**, **200''** (and expected to be closer to the source of the fume, smoke or heat emission **400**). When the third container handling vehicle **200'''** enters or has entered cell F6, the fire detection device of the third container handling vehicle **200'''** transmits data from the fire detection device to the master control system **800**. The master control system **800** processes data from the fire detection device from the first, second and third container handling vehicles **200'**, **200''**, **200'''**. If the master control system **800**, based on the processing has a reasonable prediction on the location of the source of the fume, smoke or heat **400**, no further container handling vehicles **200^x** are instructed to move closer to the expected source of the fume, smoke or heat emission **400**. Normally, arranging three container handling vehicles **200'**, **200''**, **200'''** in a triangular arrangement TA enclosing the source of the fume, smoke or heat emission **400**, will be sufficient in order to establish a reasonable prediction on location of the source for the fume, smoke or heat emission **400** is. However, if a reasonable prediction on the location of the source of the fume, smoke or heat **400** cannot be established, yet another container handling vehicle **200^x** is instructed to move to a cell closer to the expected source of the fume, smoke or heat emission **400**, which situation is indicated in FIG. 6D.

In FIG. 6D the master control system **800**, based on the input from the first, second and third container handling vehicles **200'**, **200''**, **200'''**, has instructed a fourth container handling vehicle **200''''** initially positioned in cell J1, to move to cell J3 (as illustrated by arrow AR-3) which is closer to the first, second and or third container handling vehicles **200'**, **200''**, **200'''** (and expected to be closer to the source of the fume, smoke or heat emission **400**). When the fourth container handling vehicle **200''''** enters or has entered cell J3, the fire detection device of the fourth container handling vehicle **200''''** transmits data from the fire detection device to the master control system **800**. The master control system **800** processes data from the fire detection devices from the first, second, third container handling vehicles **200'**, **200''**, **200'''**, **200''''**. A total of four container handling vehicles

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200', **200''**, **200'''**, **200''''** are arranged as a four-cornered polygon PA enclosing the source of the fume, smoke or heat emission **400**. This setup is almost certainly enough to establish a reasonable location for the source of the fume, smoke or heat emission **400**. However, in the unlikely event that four container handling vehicles **200'**, **200''**, **200'''**, **200''''** are not sufficient, the master control system **800** may instruct further container handling vehicles **200^x** with fire detection device to move closer to the expected location of the source of the fume, smoke or heat emission **400**.

In the preceding description, various aspects of an automated storage and retrieval system, vehicle and methods according to the invention have been described with reference to the illustrative embodiment. For example, in most of the figures, container handling vehicles operating on a rail system of a storage grid have been disclosed, but it is obvious that the same system and setup applies to delivery vehicles operating on a delivery rail in a delivery rail system. Thus, the description is not intended to be construed in a limiting sense. Various modifications and variations of the illustrative embodiments, as well as other embodiments of the system, which are apparent to persons skilled in the art, are deemed to lie within the scope of the present invention as defined by the following claims.

LIST OF REFERENCES

- 1 Automated storage and retrieval system
- 30 Delivery vehicle, remotely operated vehicle
- 31 Wheel arrangement delivery vehicle
- 50 Delivery rail system
- 51 First set of parallel rails, delivery rail system
- 51a, 51b Pair of rails of the first rails in delivery rail system
- 52 Second set of parallel rails, delivery rail system
- 52a, 52b Pair of rails of the second set of rails in delivery rail system
- P1 Horizontal plane of delivery rail system
- 100 Framework structure
- 102 Upright members of framework structure
- 103 Horizontal members of framework structure
- 104 Storage grid/three-dimensional grid
- 105 Storage column
- 106 Storage container
- 107 Stack
- 108 Rail system/Container handling vehicle rail system
- 110 First set of parallel rails in first direction X
- 110a, 110b Pair of rails of the first rails
- 111 Second set of parallel rails in second direction Y
- 111a, 111b Pair of rails of the second set of rails
- 112 Grid column
- 115 Grid opening
- 119 Delivery column
- 120 Delivery column
- 122 Grid cell
- 140 Delivery system
- 150 Fire detection device
- 160 Heat map
- 161 Outer portion, coarse-grain
- 162 Intermediate portion, striped black
- 163 Inner portion, solid black
- 200, 300 Container handling vehicle, remotely operated vehicle
- 200', 200'', . . . , 200^x First, second, . . . , x vehicle, remotely operated vehicle
- 201 Wheel arrangement
- 301 Wheel arrangement

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800 Master control system

X First direction

Y Second direction

P Horizontal plane of rail system

P1 Horizontal plane of delivery rail system

 W_c Width of grid cell L_c Length of grid cell W_o Width of grid opening L_o Length of grid opening

The invention claimed is:

1. An automated storage and retrieval system comprising: a master control system configured to keep track of a plurality of remotely operated vehicles operating on the automated storage and retrieval system, each of the plurality of remotely operated vehicles comprising a fire detection device configured to transmit data from the fire detection device to the master control system, wherein

the master control system comprises a processing device for processing the data from two or more of the fire detection devices, the processing of the data creating a heat map of the automated storage and retrieval system providing information in terms of a location of a fire or smoke or a current temperature or temperature profile of heat or fume emission,

wherein the master control system is configured, based on data from a first fire detection device of a first remotely operated vehicle at a first position indicating a detected heat or fume emission, to allocate a second remotely operated vehicle with a second fire detection device to move to a second position nearby the first position of the first remotely operated vehicle indicating the detected heat or fume emission to verify the detected heat or fume emission, and

wherein the master control system is configured, based on the second remotely operated vehicle providing data from the second fire detection device to the master control system indicating a presence of heat or fume emission, to:

allocate a third remotely operated vehicle with a third fire detection device to move to a third position nearby the first position of the first remotely operated vehicle and the second position of the second remotely operated vehicle;

process the data from the first fire detection device and the second fire detection device and data from the third fire detection device, and based on the processing, determine whether a reasonable prediction of a location of a source of the heat or fume emission can be given; and generate the heat map providing information in terms of a location of a fire or smoke or a current temperature or temperature profile of the heat or fume emission.

2. The automated storage and retrieval system according to claim 1, wherein the remotely operated vehicles are container handling vehicles each comprising a lifting assembly for picking up a storage container from a storage column to a position above a lowest level of a first set and a second set of wheels, wherein the lifting assembly comprises a lifting frame connectable to a storage container, and the lifting frame is configured to lift and lower the storage container from a position in the storage column to a position above a rail system of the automated storage and retrieval system.

3. The automated storage and retrieval system according to claim 1, further comprising:

a plurality of stacks of storage containers arranged in storage columns located beneath a rail system of the

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automated storage and retrieval system, wherein each storage column is located vertically below a grid opening.

4. The automated storage and retrieval system according to claim 1, wherein the remotely operated vehicles are delivery vehicles comprising a closed bottom portion and are configured to receive storage containers from an above position.

5. The automated storage and retrieval system according to claim 1, wherein a majority of the remotely operated vehicles comprises an interface connection of the fire detection devices.

6. The automated storage and retrieval system according to claim 1, further comprising a rail system comprising a first set of parallel tracks arranged in a horizontal plane and extending in a first direction, and a second set of parallel tracks arranged in the horizontal plane and extending in a second direction which is orthogonal to the first direction, wherein the first and second sets of tracks form a grid pattern in the horizontal plane comprising a plurality of adjacent grid cells, each comprising a grid opening defined by a pair of neighboring tracks of the first set of tracks and a pair of neighboring tracks of the second set of tracks; and wherein each of the plurality of remotely operated vehicles further comprises first and second sets of wheels for transport in the first direction and the second direction on the rail system.

7. A method of verifying a detected heat or fume indication on an automated storage and retrieval system comprising:

operating a master control system that keeps track of a plurality of remotely operated vehicles operating in the automated storage and retrieval system;

receiving, by the master control system, data from a first fire detection device of a first remotely operated vehicle of the plurality of remotely operated vehicles;

determining, by the master control system, that the data from the first fire detection device at a first position indicates a detected heat or fume emission;

allocating, by the master control system, a second remotely operated vehicle with a second fire detection device to move to a second position nearby the first position of the first remotely operated vehicle indicating the detected heat or fume emission to verify the detected heat or fume emission,

based on the second remotely operated vehicle providing data from the second fire detection device to the master control system indicating a presence of heat or fume emission, allocating, by the master control system, a third remotely operated vehicle with a third fire detection device to move to a third position nearby the first position of the first remotely operated vehicle and the second position of the second remotely operated vehicle; and

utilizing the master control system to:

process the data from the first fire detection device and the second fire detection device and data from the third fire detection device, and based on the processing, determine whether a reasonable prediction of a location of a source of the heat or fume emission can be given; and generate a heat map providing information in terms of a location of a fire or smoke or a current temperature or temperature profile of the heat or fume emission.

8. The method according to claim 7, wherein based on a determination that the reasonable prediction of the location of the source of the heat or fume emission can be given,

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utilizing the master control system to create a triangular arrangement (TA) enclosing the source of the heat or fume emission.

9. The method according to claim 7, wherein based on a determination that the reasonable prediction of the location of the source of the heat or fume emission cannot be given, allocating a fourth remotely operated vehicle with a fourth fire detection device to move to a fourth position nearby the first, second, and third positions of the first, second, and third remotely operated vehicles; and utilizing the master control system to process the data from the first fire detection device, the second fire detection device, and the third fire detection device, and data from the fourth fire detection device, and based on the processing determine whether a reasonable prediction of the location of a source of the heat or fume emission can be given.

10. The method according to claim 9, wherein based on a determination that the reasonable prediction of the location of the source of the heat or fume emission can be given, utilizing the master control system to create a four-cornered polygon (PA) enclosing the source of the heat or fume emission.

11. The method according to claim 7, wherein each of the first fire detection device, the second fire detection device, and the third fire detection device is a fume or smoke detector, and wherein the data from the first fire detection device, the second fire detection device, and the third fire detection device represents a presence of fume or smoke.

12. The method according to claim 7, wherein each of the first fire detection device, the second fire detection device,

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and the third fire detection device is a heat detector and wherein the data from the first fire detection device, the second fire detection device, and the third fire detection device represents a presence of heat.

13. The method according to claim 7, wherein the first fire detection device, the second fire detection device, and the third fire detection device are a combination of fume or smoke detectors and heat detectors and wherein the data from the first fire detection device, the second fire detection device, and the third fire detection device represents a presence of fume or smoke or heat.

14. The method according to claim 7, wherein the method comprises, prior to the generating of the heat map providing information in terms of the location of the fire or smoke or the current temperature or temperature profile of the heat or fume emission, using a stationary fire detection device arranged in or at the automated storage and retrieval system.

15. The method according to claim 7, wherein the automated storage and retrieval system comprises a rail system comprising a first set of parallel tracks arranged in a horizontal plane and extending in a first direction, and a second set of parallel tracks arranged in the horizontal plane and extending in a second direction which is orthogonal to the first direction, wherein the first and second sets of tracks form a grid pattern in the horizontal plane comprising a plurality of adjacent grid cells, each comprising a grid opening defined by a pair of neighboring tracks of the first set of tracks and a pair of neighboring tracks of the second set of tracks.

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